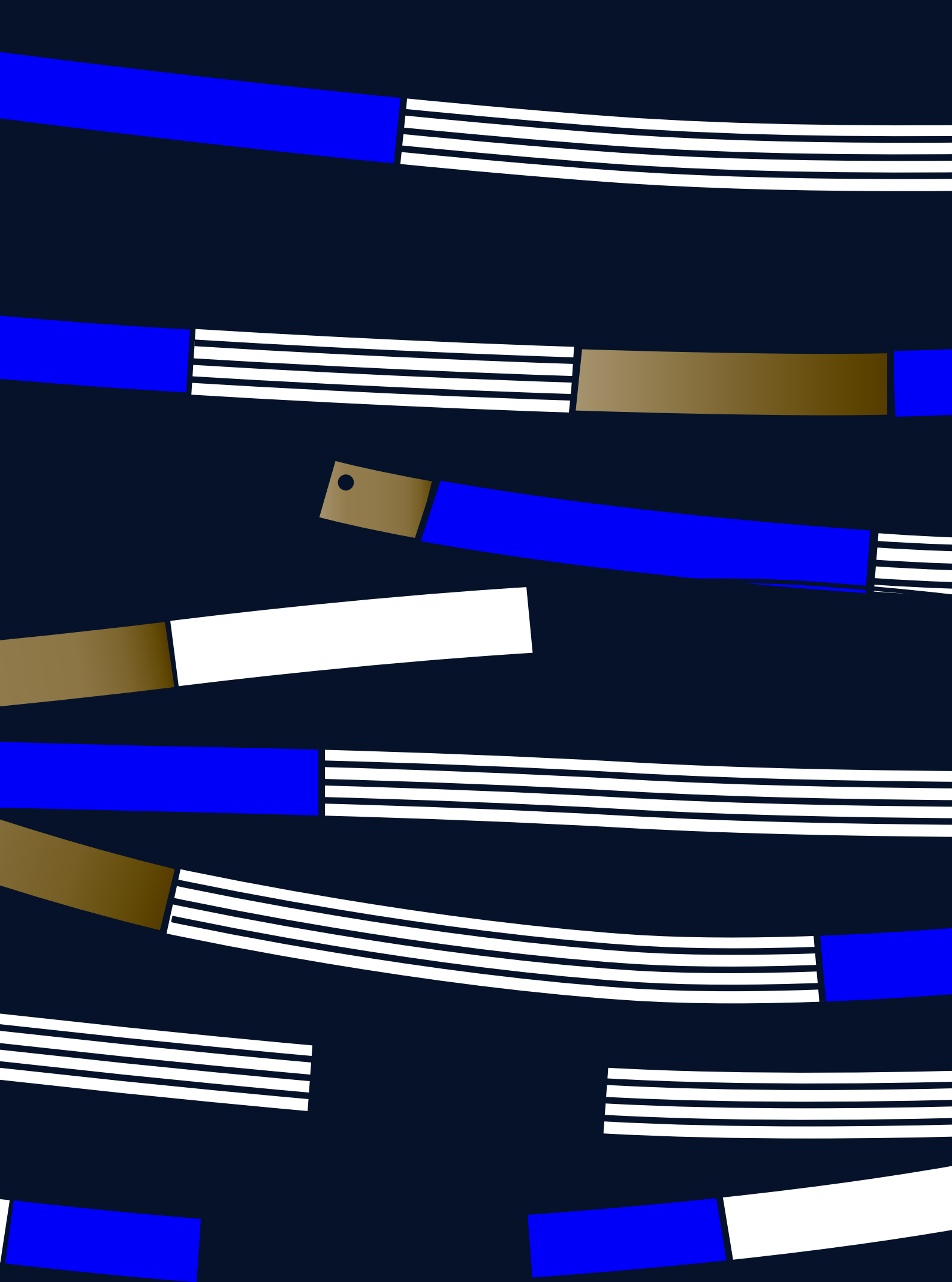




Chapter 7 - Waves

=> **Wave motion** - energy transferred through moving oscillations.

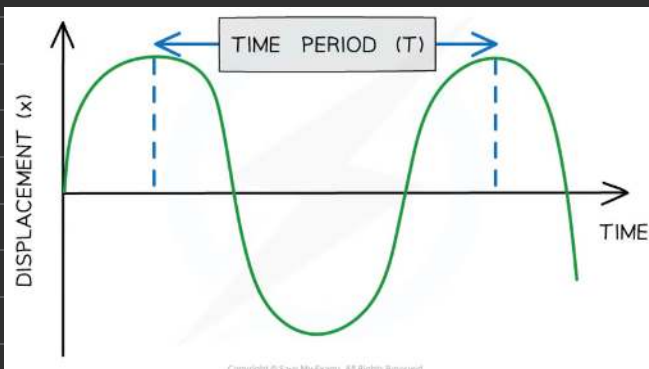
=> When vibrations are _____ to direction of wave travel,

- **perpendicular, transverse** wave is formed.
- **parallel, longitudinal** wave is formed.

+ **Displacement** - the distance from the equilibrium position.

+ **Amplitude** - maximum displacement of a particle.

+ **Wavelength** - Minimum distance between two points in the same phase.



=> **Frequency (Hz)** - number of complete oscillations per unit time.

$$f = 1/T \text{ (Time period)}$$

=> **Phase difference** - difference in position of the crests or troughs of two different waves of the same frequency.

It can be measured in fraction of a wavelength, degrees or radians.

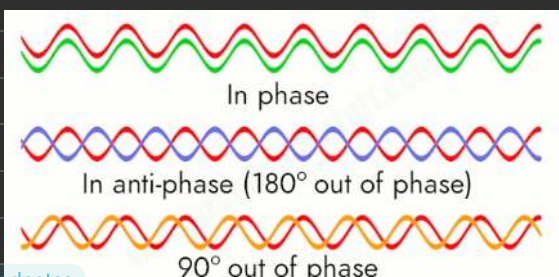
+ **In phase** - when crests and troughs are aligned.

- has a fixed difference of **360** degrees or 2π .

+ **Anti phase** - when crest of one wave aligns with the trough of another wave.

- has a fixed difference of **180** degrees or π .

+ **Out phase** - Any degree between 0 and 360 degrees



=> **Progressive waves** - transfer energy between points.

=> **Stationary waves** - don't transfer energy between points.

=> **Cathode-Ray Oscilloscope** displays and measure waveforms of electrical circuits.

- AC current is represented as transverse waves.

- x-axis shows time and y-axis shows voltage.

- Period of the wave can be determined from time-base, it shows how many seconds each division represents.

=> **Time period (time taken for one complete oscillation) = Total time / No. of wavelengths**

=> Deriving wave speed formula

$$\text{wave speed} = \text{distance} / \text{time}$$

$$= \lambda / T$$

$$\text{frequency} = 1/T$$

$$\text{hence, } v = \lambda / (1/T) = f \lambda$$

=> **For a wave at constant speed,**

$\Leftrightarrow$ **wavelength decreases, frequency increases**

=> For some questions, **c** (speed of light - 3×10^8) can be used instead of **v** (speed for ANY wave).

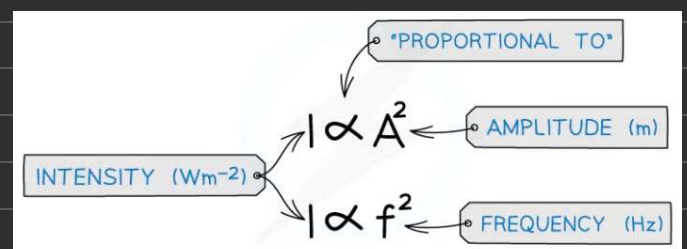
- As progressive waves transfer energy, the amount of energy passing through a unit area per unit time is the **intensity of the wave**.

$$I = P / A$$

- Intensity (Wm^{-2})

- Power (W)

- Area (m^2)



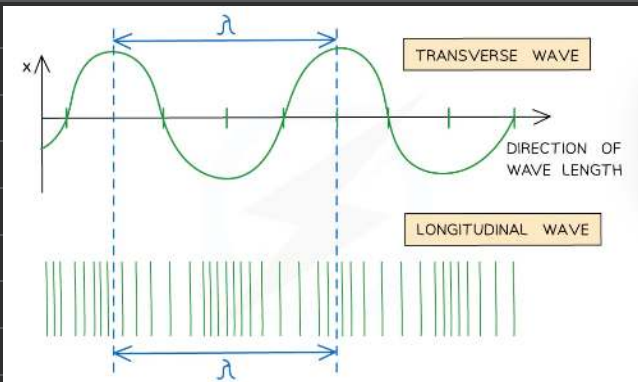
=> If frequency or amplitude doubles, intensity increases by a factor of 4 (2^2).

Transverse wave - particles oscillate perpendicular to the direction of wave travel.

- shows crests and troughs
- Examples: electromagnetic waves and vibration
- can be polarised
- can be shown using ropes

Longitudinal waves - particles oscillate parallel to the direction of wave travel.

- shows compressions and rarefactions
- Examples: sound waves and ultra sound waves
- cannot be polarised
- can be shown using slinky springs



=> **Doppler effect** - frequency change due to relative motion between a source and an observer.

- When the observer and source are **both stationary**, the waves are at the **same frequency and wavelength** for both.
- When the source moves **towards the observer**, the **wavelength is shorter** and **frequency is higher**.

$$f_o = f_s \left(\frac{V}{V \pm V_s} \right)$$

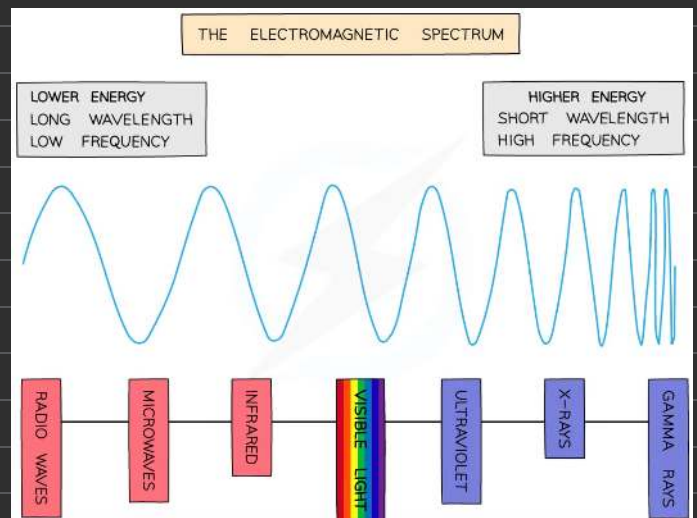
Labels in the diagram: SOURCE FREQUENCY (Hz) points to f_s ; OBSERVED FREQUENCY (Hz) points to f_o ; WAVE VELOCITY (ms^{-1}) points to V ; SOURCE VELOCITY (ms^{-1}) points to V_s .

- if source is moving **towards**, the denominator is **subtract**
 - source is moving **away**, the denominator **add**
- => Wave velocity for sound waves is 340ms^{-1} .
- => **Observer** is always **stationary** and the **source** is always the object which is **moving**.

Properties of electromagnetic waves:

- transverse waves
- all travel in vacuum..
- travel at the same speed in vacuum (3.8×10^8)
- Since they are transverse, all waves in this spectrum can be reflected, refracted, diffracted, polarised and produce interference patterns.

WAVE	USE
RADIO	• COMMUNICATION (RADIO AND TV)
MICROWAVE	• HEATING FOOD • COMMUNICATION (WIFI, MOBILE PHONES, SATELLITES)
INFRARED	• REMOTE CONTROLS • FIBRE OPTIC COMMUNICATION • THERMAL IMAGING (MEDICINE AND INDUSTRY) • NIGHT VISION • HEATING OR COOKING THINGS • MOTION SENSORS (FOR SECURITY ALARMS)
VISIBLE LIGHT	• SEEING AND TAKING PHOTOGRAPHS/VIDEOS
ULTRAVIOLET	• SECURITY MARKING (FLUORESCENCE) • FLUORESCENT BULBS • GETTING A SUNTAN.
X-RAYS	• X-RAY IMAGES (MEDICINE, AIRPORT SECURITY AND INDUSTRY)
GAMMA RAYS	• STERILISING MEDICAL INSTRUMENTS • TREATING CANCER

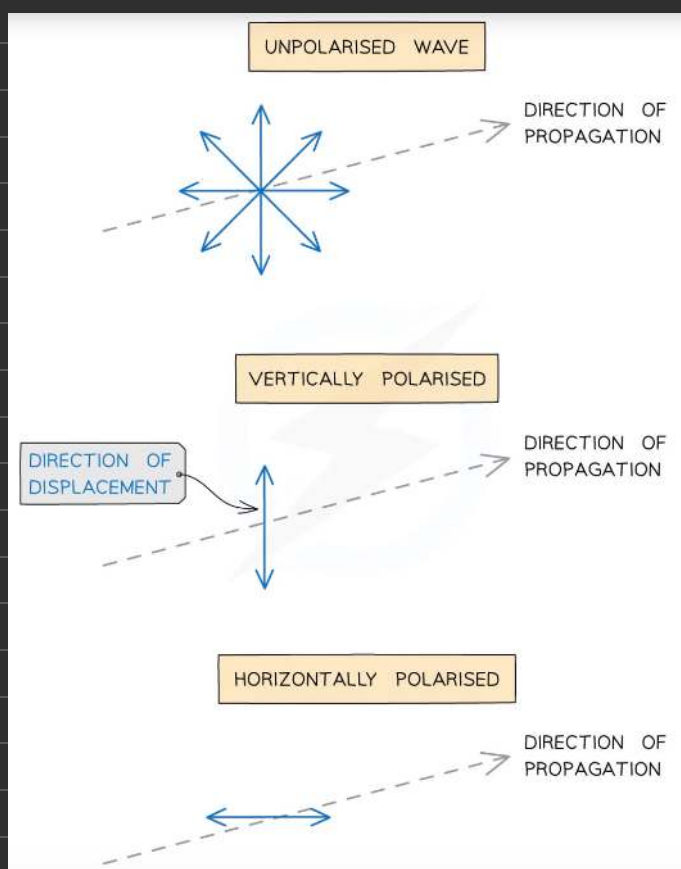


The higher the frequency, the higher the energy of radiation (highly ionizing and harmful).

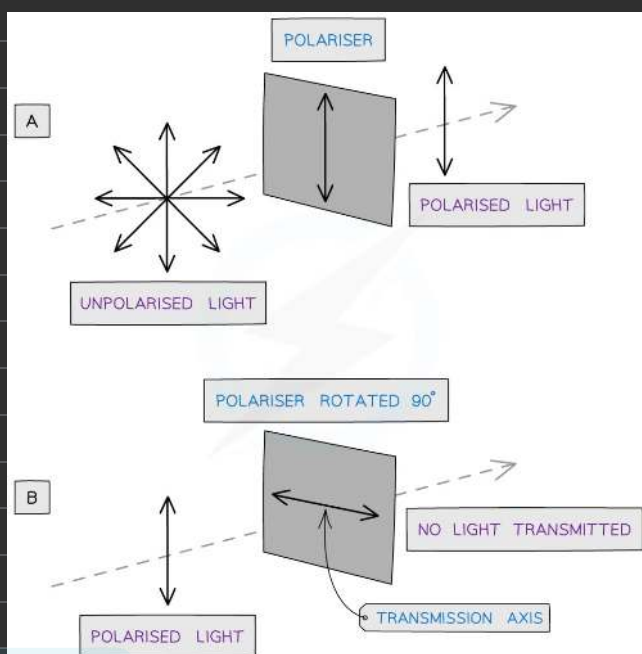
Radiation	Approximate wavelength range / m
Radio	> 0.1
Microwaves	$0.1 - 1 \times 10^{-3}$
Infra-red	$1 \times 10^{-3} - 7 \times 10^{-7}$
Visible	$4 \times 10^{-7} - 7 \times 10^{-7}$
Ultra-violet	$4 \times 10^{-7} - 1 \times 10^{-8}$
X-rays	$1 \times 10^{-8} - 4 \times 10^{-13}$
Gamma rays	$1 \times 10^{-10} - 1 \times 10^{-16}$

range of frequencies can be found using $c = f \lambda$ where c is $3.0 \times 10^8 \text{ m/s}$

- **Visible light** - Range of wavelengths (400-700nm) which is visible to humans.
- **Polarization** - Oscillations in one direction, perpendicular to the direction of propagation.



- Longitudinal waves cannot be polarized as they oscillate **parallel** to the direction of travel.
- Waves can be polarized using **polarizer** or **polarizing filter** which only allows oscillations in a certain plane to be transmitted.



=> Light can also be polarized through **reflection, refraction and scattering** (not diffraction).

- Example of polarization: polaroid sunglasses which reduce the glare caused by sunlight for a driver to see through windows.

=> When intensity I_0 of unpolarized light passes through a polarizer, the intensity of the transmitted polarized light **falls by half**.

=> **Unpolarized light goes through two filters:**

- polariser
- analyzer
 - if the analyzer has the same orientation as the polarizer, the light transmitted by the analyzer has the same intensity as the light incident on it
 - if they have different orientation, we use **Malus's law**:

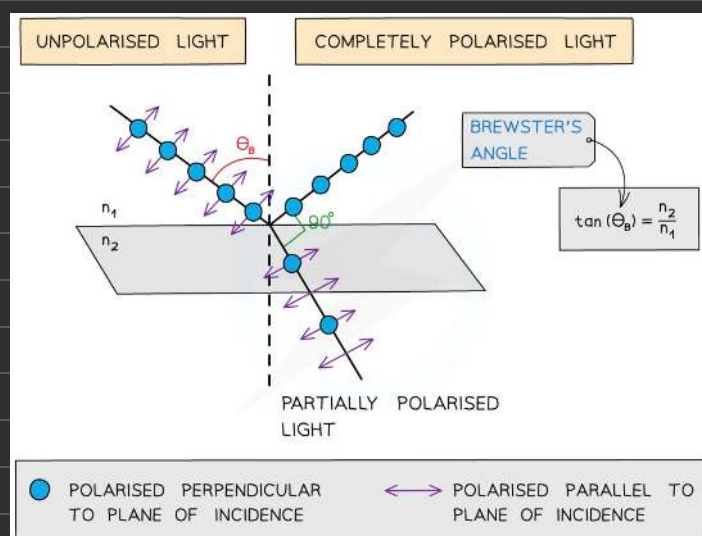
$$I = I_0 \cos^2(\Theta)$$

Where:

- I : INTENSITY OF TRANSMITTED LIGHT (Wm^{-2})
- I_0 : MAXIMUM INTENSITY (Wm^{-2})
- Θ : ANGLE BETWEEN POLARISED LIGHT AND TRANSMISSION AXIS (DEGREES OR RADIAN)

=> **Half rule** - When unpolarized light passes through the first polarizer, half the intensity of the wave is always lost ($\frac{I_0}{2}$).

=> **Brewster's angle** - an angle of incidence at which light with a particular polarisation is perfectly transmitted through a surface.

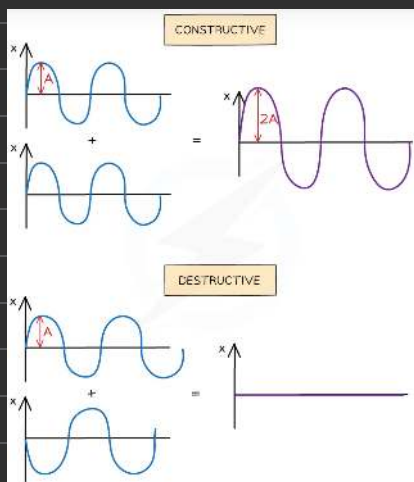


Chapter 8 - Superposition

Principal of superposition - When two or more waves meet, the resultant displacement is the sum of the displacement of the individual waves.

=> When two waves with same frequency and amplitude meet at a point, they superpose either:

- **in phase** - constructive interference, the peaks and troughs line up, hence the amplitude of resultant wave doubles.
- **anti-phase** - destructive interference, the peak of one wave lines up with trough of the other, hence the resultant wave has no amplitude.



The **principle of superposition applies to all types of waves** like transverse and longitudinal.

=> Stationary waves are produced by the superposition of two waves of the same frequency and amplitude traveling in opposite directions.

- Vibrations caused by stationary waves on a stretched string produce sound, like guitars.

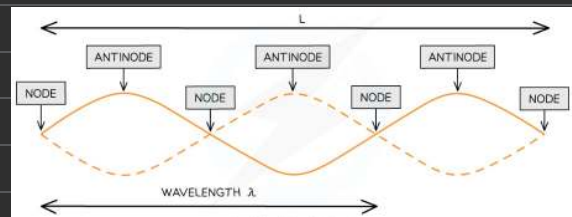
=> Stationary wave on a stretched string - As the frequency of the oscillator changes, standing waves with different numbers of minima (nodes) and maxima (antinodes) form.

=> A microwave source is placed in line with a reflecting plate and a small detector between the two. By moving the detector, it can pick up the minima (nodes) and maxima (antinodes) of the stationary wave pattern.

Air column - stationary waves can be produced by sound wave.

=> Formation of stationary waves

- nodes - no vibration
 - they are fixed, doesn't move.
 - antinodes - vibrations are at maximum amplitude
 - moves in vertical direction only
- + Between nodes, all points on the stationary are in phase



- L is the length of the string
 - 1 wavelength λ is only a portion of the length of the string
- => When a stationary wave, such as a vibrating string, is fixed at both ends, a single loop made up of two nodes and an antinode, known as **first harmonic or fundamental mode of vibration**.
- Frequency increases, the higher harmonics begin to appear.

=> **nth harmonic has n antinodes and n + 1 nodes**

Air column fundamental wave	Length L / m	Resonant frequencies f / Hz	Value of n
	$L = \frac{n\lambda}{2}$	$f = \frac{nv}{2L}$	$n = 1, 2, 3$
	$L = \frac{n\lambda}{4}$	$f = \frac{nv}{4L}$	$n = \text{odd}$
	$L = \frac{n\lambda}{2}$	$f = \frac{nv}{2L}$	$n = 1, 2, 3...$

Diffraction - Spreading out of waves when they pass an obstruction.

- most noticeable when the width of the gap is equal to wavelength.
- represented by wavefronts.
- amplitude changes as some energy is dissipated when a wave is diffracted through a gap.

=> if the gap is larger than wavelength, the waves are no longer spread out.

=> Ripple tanks are used to demonstrate diffraction of waves.

⇒ **higher frequency, shorter wavelength**

Interference - when two or more waves meet, the resultant displacement is the sum of the displacement of the individual waves.

=> two types of interference

- **in phase** - constructive interference, the peaks and troughs line up, hence the amplitude of resultant wave doubles.
- **anti-phase** - destructive interference, the peak of one wave lines up with trough of the other, hence the resultant wave has no amplitude.
- if both waves aren't in phase nor anti-phase, the resultant amplitude is between both.

=> Coherent - when waves have same frequency and constant phase difference.

- for example, laser light.

Interference of sound, light and microwaves can be demonstrated with slits or diffraction gratings

	Constructive	Destructive
Sound	When two compressions or two rarefactions line up, and sound appears louder.	When a compression lines up with rarefaction, the sound is quieter.
Microwave	Regions where the detector picks up a maximum amplitude.	Regions where the detector picks up no signal.
Light waves	Bright fringes, has highest intensity in the middle.	Dark fringes, has zero intensity.

=> For two-source interference fringes to be observed, the sources of the wave must be:

- Coherent (constant phase difference)
- Monochromatic (single wavelength)

=> Path difference - difference in distance that two waves travel from their point of intersection.

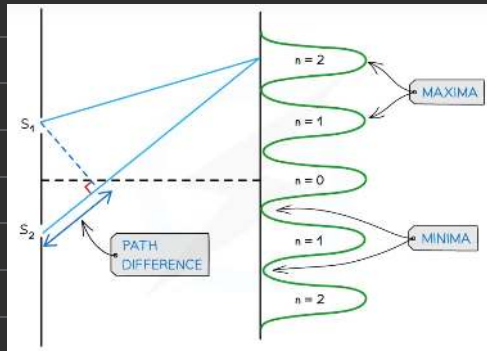
CONSTRUCTIVE INTERFERENCE: PATH DIFFERENCE = $n\lambda$	DESTRUCTIVE INTERFERENCE: PATH DIFFERENCE = $(n + \frac{1}{2})\lambda$
---	--

where $n = 0, 1, 2, 3...$

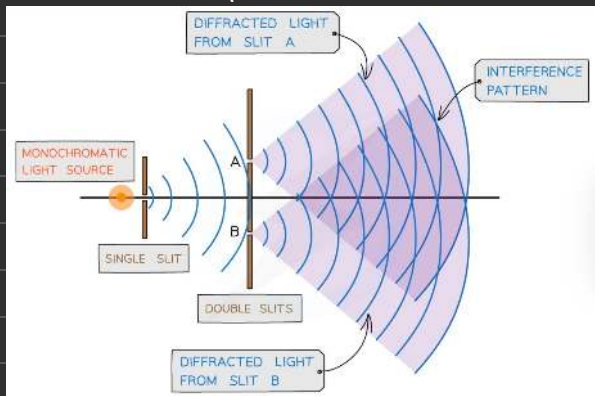
=> For constructive interference (maxima), the difference in wavelength will be integer number of whole wavelengths.

=> For destructive interference (minima), the difference in wavelength will be integer number of whole wavelengths plus a half wavelength.

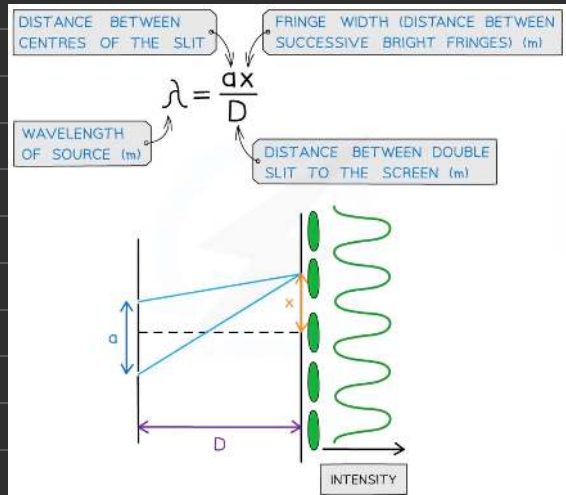
- n is the order of the maxima/minima



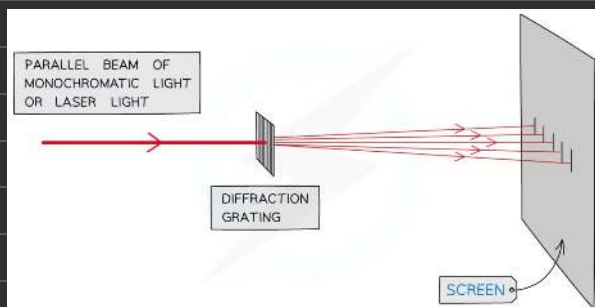
Young's double slit experiment demonstrates how light waves produced an interference pattern.



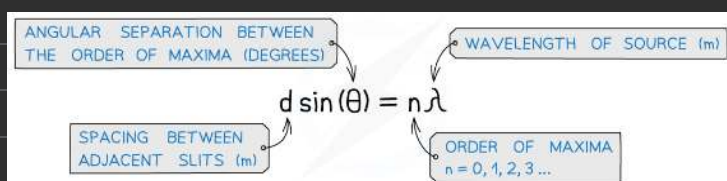
=> When a monochromatic light source is placed behind a single slit, the light is diffracted producing two light sources at the double slits A and B. Since both light sources originate from the same primary source, they are coherent and will therefore create an observable interference pattern made up of bright and dark fringes.



=> A diffraction grating is a plate on which there is a very large number of parallel, identical, close-spaced slits, when monochromatic light is incident on a grating, a pattern of narrow bright fringes is produced on a screen.



Diffraction grating equation for the angle of bright fringes:



+ Lines per m (or per mm, per nm...) on the grating is represented by the symbol N .

$$d = 1/N$$

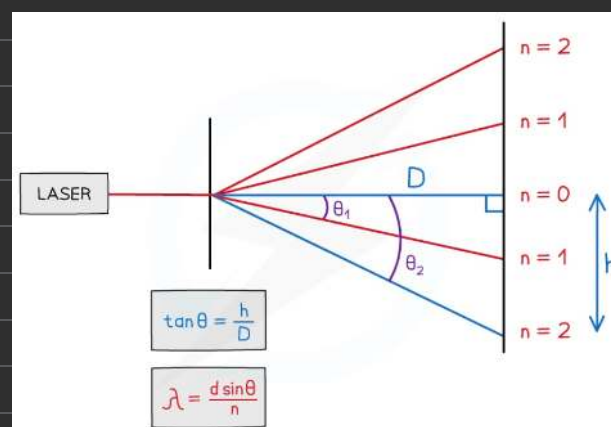
=> Angular separation - the angle is taken from the centre and can be calculated by rearranging the grating equation.

- The angular separation between two angles is found by subtracting the smaller angle from the larger one.

=> Highest order of maxima can be calculated by:

$$n = \frac{d}{\lambda}$$

- n must be an integer value and should be rounded down if the value is in decimal.



Chapter 9 - Electricity

=> **Electric current** - flow of charge carriers

- units: **amperes (A)**, measured using: ammeter
- ammeters** should be connected in **series**

=> Electrons are negatively charged; they flow towards positive terminal FROM negative terminal.

=> **Conventional current** - flow of positive charge from positive terminal to negative terminal.

=> **Quantization** - Charge always exists in integral multiple of fundamental charge ($1.6 \times 10^{-19} \text{C}$).

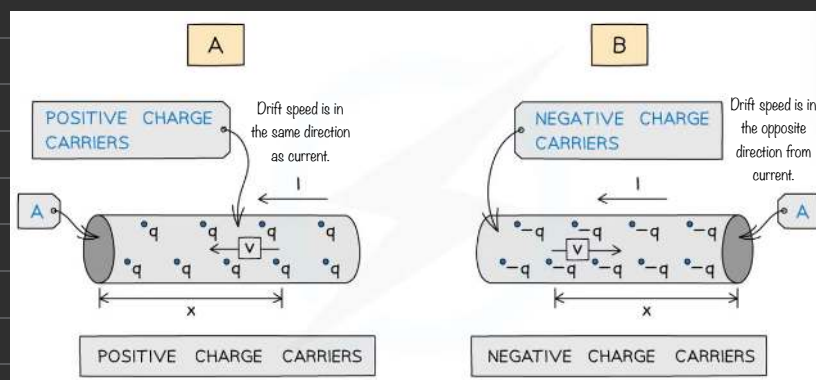
=> **Current (I)** - charge passing through a circuit per unit time.

=> **Charge (Q)** unit - coulombs (C)

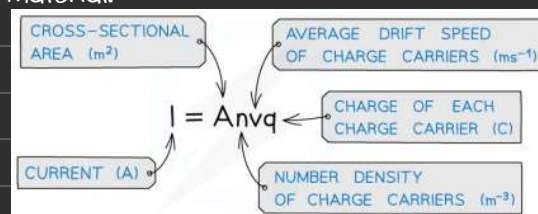
$$Q = It$$

- In conductors, the charge carrier is usually free electrons.

=> **Drift speed** - average speed of charged particles when there is current in a conductor.

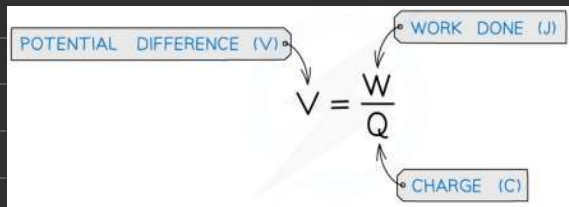


=> **Number density** - number of charge carriers per unit volume in a material.



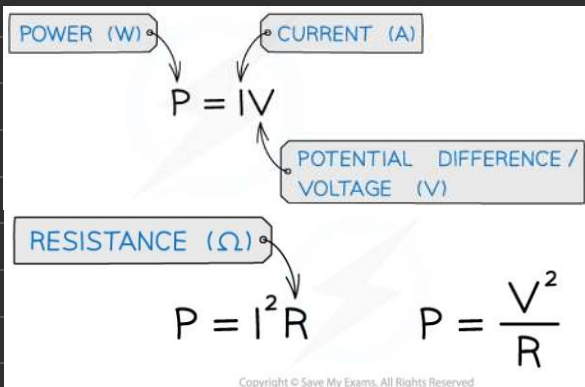
=> **Potential difference** - energy transferred per unit charge flowing from one point to another.

- measured in **volts** {Joule per coulomb (J C^{-1})} using **voltmeter** (It is always set up in **parallel**)



Potential difference - work done per unit charge.

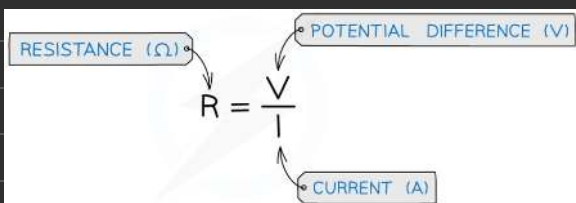
Current - rate of flow of charge



- if current or voltage doubles, power will be four times greater.

=> Resistance is the opposition to current

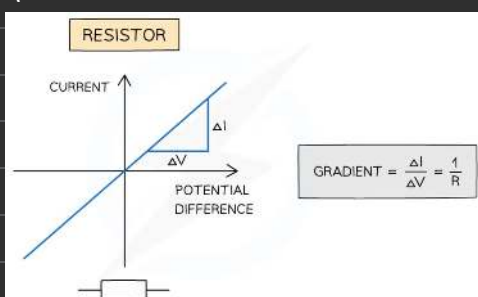
- For a given potential difference: The **higher the resistance, the lower the current.**



- Resistance** is measured in **Ohms (Ω)** {one volt per ampere}

=> **Higher resistance, smaller current.**

=> **Ohm's law** - for a conductor at constant temperature, the current through it is proportional to the potential difference across it.

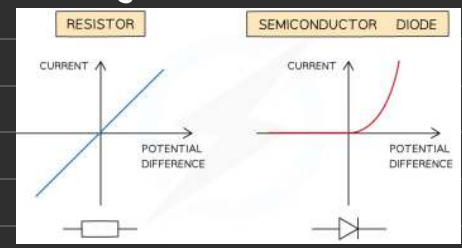


=> If the graph is a straight line through the origin:

- A resistor obeys Ohm's law
- A filament lamp does not obey Ohm's law

=> As potential difference (voltage) across a component is increased, the current also increases (by Ohm's law).

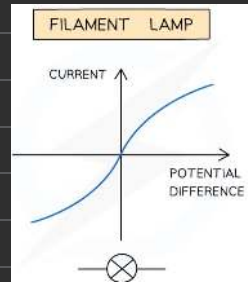
=> **Diode** - used in a circuit to allow current to flow only in a specific direction:



- Forward bias** - current is in direction of arrowhead symbol; sharp increase in potential difference and current.

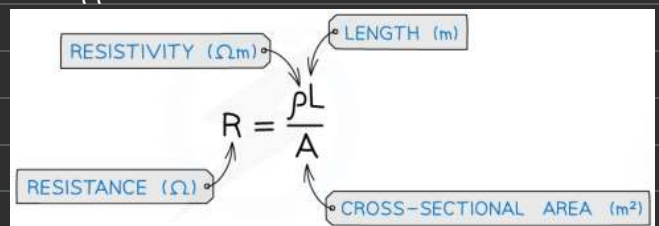
- Reverse bias** - diode is switched around; zero reading of current or potential difference.

The I-V graph for a filament lamp shows the current increasing at a proportionally slower rate than the potential difference.



=> This is because:

- as current increases, temperature of filament lamp increases.
- increase in temperature increases resistance.
- resistance opposes current, thus current decreases.

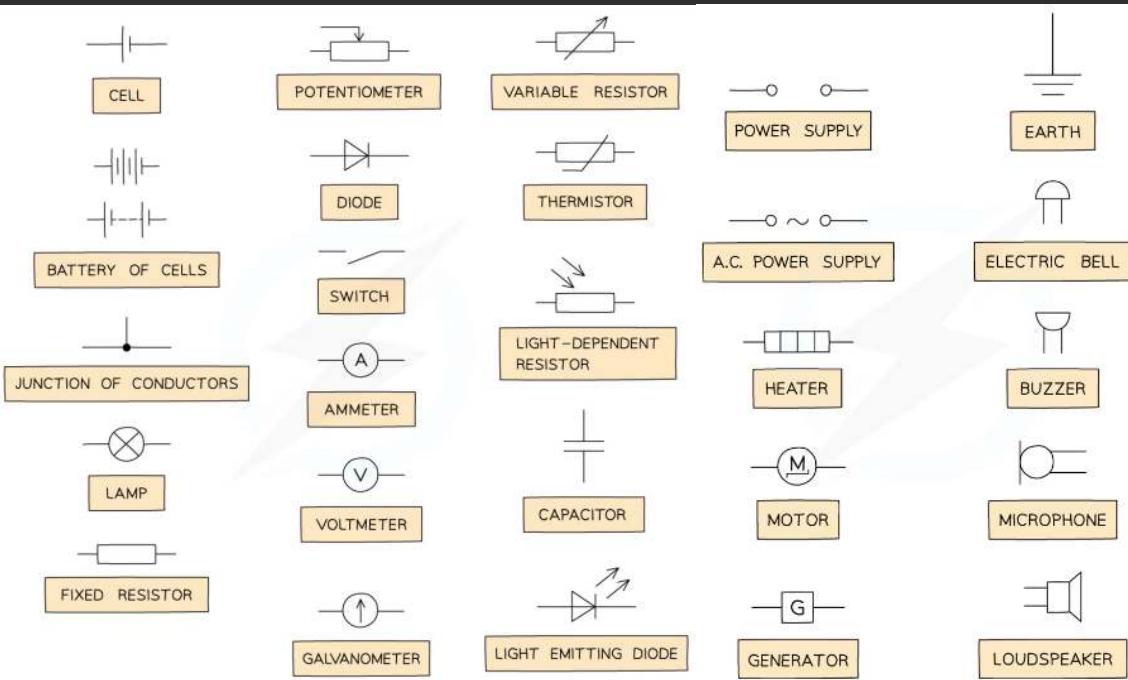


- the longer the wire, the greater its resistance
- the thicker the wire, the smaller its resistance
- if the diameter doubles, the area quadruples causing the resistance to drop by a quarter.

=> Non-ohmic conductor and sensory resistors:

- Light-dependent resistor (LDR)** - as the **light intensity increases**, the **resistance of an LDR decreases.**
- Thermistor** - as **temperature increases**, **resistance decreases.**

Chapter 10 - D.C. Circuits



=> Internal resistance: resistance of a cell which cause charge to dissipate some electrical energy from the power supply itself. It cause loss of voltage.

=> Terminal potential difference (R): voltage available in the circuit itself.

=> Lost volts (r): voltage lost in the cell due to internal resistance.

• Lost volts = e.m.f - terminal p.d

=> e.m.f is the total, or maximum, voltage available to the circuit.

=> The function of the most common components are:

- **Switch:** Turn the circuit on (closed), or off (open)
- **Fixed resistor:** A fixed resistance that limits the flow of current.
- **Variable resistor:** A resistor with slider that can be used to change its resistance. Used often in dimmer switches and volume controls.
- **Thermistor**
- **Light-dependent resistor (LDR)**
- **Diode:** used to convert AC to DC current
- **Light-emitting diode (LED):** emits light when a current passes through it.
- **Ammeter:** Used to measure the current in a circuit.
- **Voltmeter:** Use to measure the potential difference.

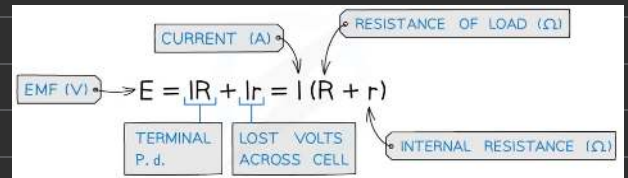
Electromotive force (e.m.f) - energy transferred per unit charge in driving a charge around a complete circuit.

$$\text{E.M.F.} = \frac{\text{ENERGY TRANSFERRED FROM OTHER FORMS TO ELECTRICAL}}{\text{CHARGE}}$$

=> e.m.f is also the potential difference across the cell when no current is flowing.

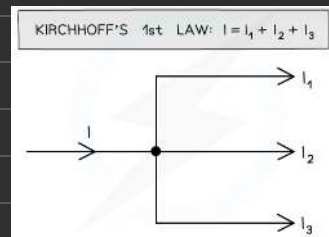
=> e.m.f can be measured by connecting a high-resistance voltmeter around the terminals of the cell in an open circuit.

$$\text{P.D.} = \frac{\text{ENERGY TRANSFERRED FROM ELECTRICAL TO OTHER FORMS}}{\text{CHARGE}}$$



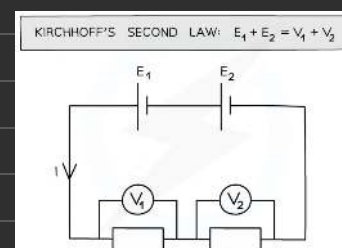
=> **Kirchhoff's first law** - The sum of the current **entering** a junction always **equal** the sum of the currents **out** of the junction.

- junction - where three circuit paths meet
- branch - path connecting two junctions

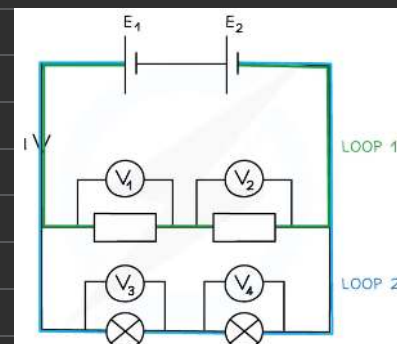


=> **Kirchhoff's second law** -

- The sum of the e.m.f's in a closed circuit equals the sum of the potential differences.



=> A closed circuit loop acts as its own independent series circuit and each one separates at a junction. A parallel circuit is made up of two or more of these loops.



=> If one light breaks, voltage and current can still flow through for the rest of the lights and appliances.

=> In a series circuit, the combined resistance of two or more resistors is the sum of the individual resistances

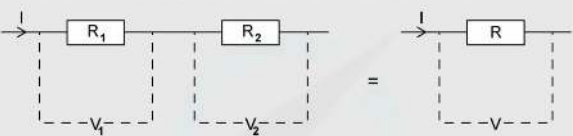
=> In a series circuit:

- The current is the same through all resistors
- The potential difference is split between all the resistors

=> Derivation for resistors in series:

CONSIDER TWO RESISTORS R_1 AND R_2 CONNECTED IN SERIES. A SINGLE RESISTOR R IS EQUIVALENT TO THEM.

FROM KIRCHHOFF'S FIRST LAW: THE CURRENT I THROUGH EACH RESISTOR IS THE SAME SINCE THEY'RE CONNECTED IN SERIES (NO JUNCTIONS)



FROM KIRCHHOFF'S SECOND LAW: THE TOTAL p.d OF BOTH RESISTORS IN A CLOSED CIRCUIT LOOP MUST EQUAL THE SUM OF THE p.ds (THE p.d ACROSS THE SINGLE RESISTOR)

$$V = V_1 + V_2$$

FROM OHM'S LAW, POTENTIAL DIFFERENCE IS GIVEN BY THE PRODUCT OF CURRENT AND RESISTANCE

$$IR = IR_1 + IR_2$$

SINCE CURRENT I IS THE SAME FOR ALL RESISTORS, DIVIDING BY I

$$R = R_1 + R_2$$

THIS EQUATION CAN BE EXTENDED SO THE EQUIVALENT RESISTOR R OF SEVERAL RESISTORS CONNECTED IN SERIES IS GIVEN BY

$$R = R_1 + R_2 + R_3 + R_4 \dots$$

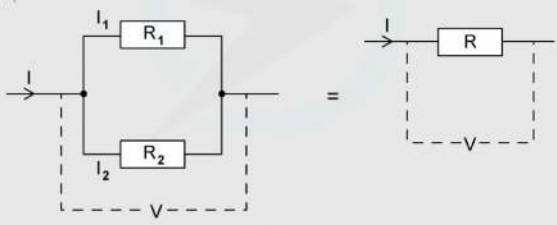
=> In a parallel circuit, the reciprocal of the combined resistance of two or more resistors is the sum of the reciprocal of the individual resistances

=> In a parallel circuit:

- The current is the split at the junction (and therefore between resistors)
- The potential difference is the same through all resistors

CONSIDER TWO RESISTORS R_1 AND R_2 CONNECTED IN PARALLEL. A SINGLE RESISTOR R IS EQUIVALENT TO THEM.

FROM KIRCHHOFF'S FIRST LAW: THE CURRENT THROUGH EACH RESISTOR WILL BE DIFFERENT DUE TO THE CURRENT SPLITTING AT THE JUNCTION. THE CURRENT THROUGH THE EQUIVALENT RESISTOR R WILL BE THE TOTAL CURRENT I



FROM KIRCHHOFF'S SECOND LAW, THE P.D ACROSS EACH RESISTOR IN DIFFERENT BRANCHES IS THE SAME. THE RESISTOR R WILL HAVE THAT SAME p.d ACROSS IT TOO

$$V = V_1 = V_2$$

REARRANGING OHM'S LAW $V = IR$ FOR CURRENT I

$$I = \frac{V}{R}$$

$$\frac{V}{R} = \frac{V}{R_1} + \frac{V}{R_2}$$

SINCE POTENTIAL DIFFERENCE V IS THE SAME FOR ALL RESISTORS, DIVIDING BY V

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

THIS EQUATION CAN BE EXTENDED SO THE EQUIVALENT RESISTOR R OF SEVERAL RESISTORS CONNECTED IN PARALLEL IS GIVEN BY

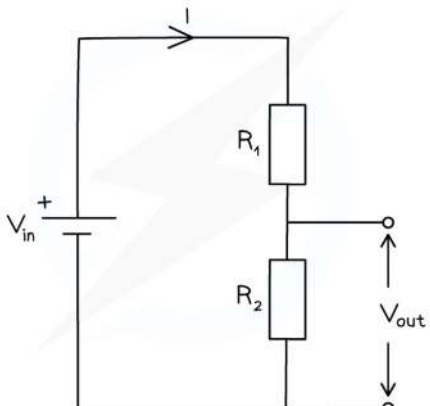
$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \dots$$

=> **Potential dividers** are circuits which produce an output voltage as a fraction of its input voltage

+ it has three main purposes:

- provide a variable potential difference
- enable a specific potential difference to be chosen
- split the potential difference of a power source between two or more components

POTENTIAL DIVIDER EQUATION: $V_{out} = \frac{R_2}{R_1 + R_2} V_{in}$

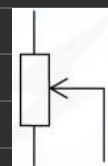


=> **Potentiometer** - a variable resistor connected as a potential divider to give a continuously variable output voltage. It consists of a coil of wire with a sliding contact.

=> **Galvanometer** - type of sensitive ammeter used to detect electric current. It is used in potentiometer to measure e.m.f. between two points in a circuit. It is made from a coil of wire wrapped around an iron core that rotates inside a magnetic field.

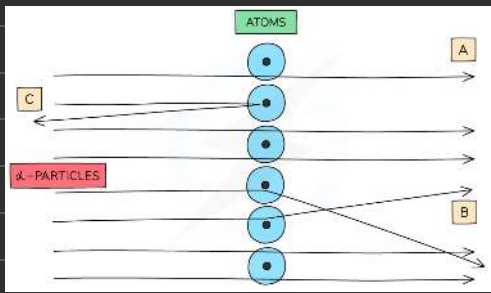
- **null deflection** - when the arrow is facing directly upwards, there is no current.
- A galvanometer has p.d of zero when the potential on one side equals the potential on the other side.

=> **The longer the length of a wire, the higher the p.d.**



Chapter 11 - Particle physics

=> α -particles are the nucleus of a helium atom and are positively charged.



- (A) majority of α -particles went straight through, which suggests **atom is mainly empty space**.
- (B) some α -particles deflected through small angles $< 10^\circ$, which suggest **positive nucleus is at the centre**.
- (C) only a small number of α -particles deflected straight back which suggests that the **nucleus is extremely small and this is where the mass and charge of the atom is concentrated**.

- Rutherford's experiment concluded that atoms consist of small dense positively charged nuclei, surrounded by negatively charged electrons.

=> Atom consists of:

PARTICLE	RELATIVE CHARGE	RELATIVE MASS
PROTON	+1	1
NEUTRON	0	1
ELECTRON	-1	1/2000 (NEGLECTIBLE)

=> All matter particles have antimatter counterparts.

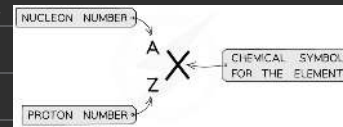
- antimatter** - material made up of antiparticles which are identical to their matter counterpart but with the opposite

MATTER	CHARGE	ANTIMATTER	CHARGE
ELECTRON e^-	-1	POSITRON e^+	+1
PROTON p	+1	ANTI-PROTON $\bar{p}$	-1
NEUTRON n	0	ANTI-NEUTRON $\bar{n}$	0

- A neutral particle, such as a neutron or neutrino, is its own antiparticle.

=> Unified atomic mass unit (u) = 1.66×10^{-27} kg

=> **Isotopes** - elements having same number of protons, but different number of neutrons.



Particle	Mass / u
Proton	1
Neutron	1
Electron	0.0005
Alpha (α)	4

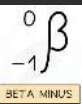
- Since isotopes have an **imbalance of neutrons and protons**, they are **unstable**. This means they constantly **decay and emit radiation to achieve a more stable form**.
- Nucleon number (A)** = total number of protons and neutrons in the nucleus

=> **Constituents** - numbers of protons and neutrons in an atom.

=> In nuclear equations, the **nucleon number** and **charge** are always **conserved**.

=> Some elements in the nuclei are **unstable**. In order to become stable, they **emit particles** or electromagnetic radiation. These nuclei are **radioactive**.

- Alpha (α) particles** have high energy particles.
- Beta (β^-) particles** are high energy electrons emitted from the nucleus. They are emitted by nuclei that have too many neutrons.
 - it is moderately ionizing as they have charge of $+1e$.
 - it is able to do some slight damage to cells (less than alpha but more than gamma).
 - it is moderately penetrating.
 - it can be stopped by a few millimetres of aluminium foil.



BETA MINUS

- Gamma (γ) rays** are high energy electromagnetic waves.
 - they are emitted by nuclei that need to lose some energy
 - If these particles hit other atoms, they can knock out electrons, ionizing the atom



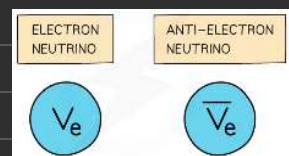
Particle	Composition	Mass / u	Charge / e	Speed / c
Alpha (α)	2 protons + 2 neutrons	4	+2	0.05
Beta minus (β^-)	Electron (e^-)	0.0005	-1	> 0.99
Beta plus (β^+)	Positron (e^+)	0.0005	+1	> 0.99
Gamma (γ)	Electromagnetic wave	0	0	1

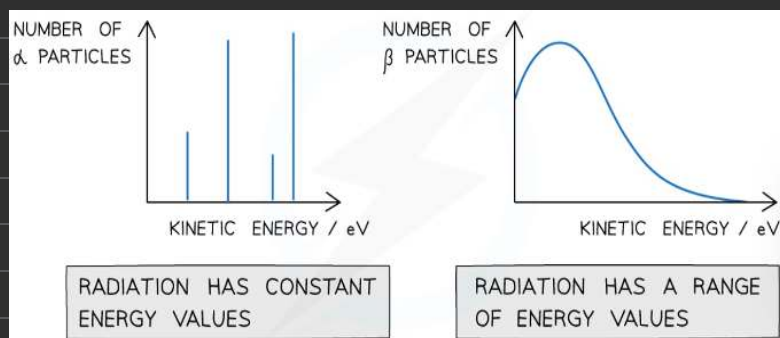
- u is the atomic mass unit (see "Atomic Mass Unit (u)")
- e is the charge of the electron: 1.60×10^{-19} C
- c is the speed of light: 3×10^8 m s $^{-1}$

=> **Electron neutrino** - type of subatomic particle with no charge and negligible mass.

=> The anti-neutrino is the antiparticle of a neutrino

- Electron **anti-neutrinos** are produced during **β^- decay**
- Electron **neutrinos** are produced during **β^+ decay**





+ Alpha particles have **discrete energy levels**.

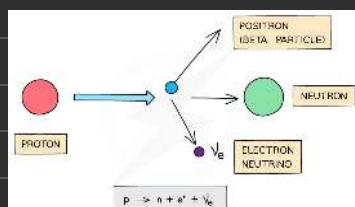
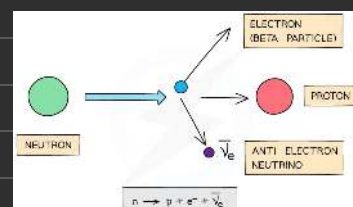
+ β particles have a **continuous range** of energies as energy released in beta decay is shared. Between beta particles and neutrinos and anti-neutrinos.

=> **Alpha (α) decay** involves a nucleus emitting an alpha particle and decaying into a different nucleus.

- mostly occurs in large, unstable nuclei with many protons.

=> **Beta (β^-) decay** is when a neutron turns into a proton emitting an electron and an anti-electron neutrino.

=> **Beta (β^+) decay** is when a proton turns into a neutron emitting a positron (anti-electron) and an electron neutrino.



=> **Quarks** are fundamental particles that make up **Hadrons (protons and neutrons)**

- **fundamental** - quarks that are not made up of any other particle such as electron.
- **quarks always exists in pairs or groups of three:**

u $+\frac{2}{3}e$ UP	c $+\frac{2}{3}e$ CHARM	t $+\frac{2}{3}e$ TOP
d $-\frac{1}{3}e$ DOWN	s $-\frac{1}{3}e$ STRANGE	b $-\frac{1}{3}e$ BOTTOM
$\bar{u}$ $-\frac{2}{3}e$ ANTI-UP	$\bar{c}$ $-\frac{2}{3}e$ ANTI-CHARM	$\bar{t}$ $-\frac{2}{3}e$ ANTI-TOP
$\bar{d}$ $+\frac{1}{3}e$ ANTI-DOWN	$\bar{s}$ $+\frac{1}{3}e$ ANTI-STRANGE	$\bar{b}$ $+\frac{1}{3}e$ ANTI-BOTTOM

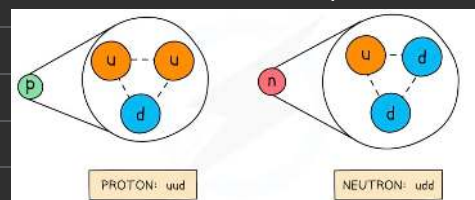
e.g. ANTI-PROTON: $\bar{p}$ IS MADE UP OF QUARKS: $\bar{u} \bar{u} \bar{d}$

ANTI-NEUTRON: $\bar{n}$ IS MADE UP OF QUARKS: $\bar{u} \bar{d} \bar{d}$

=> **Leptons** are a group of fundamental particles, they are not made up of any other particles.

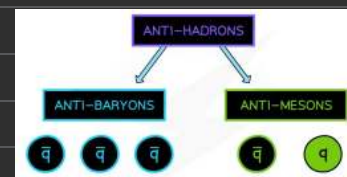
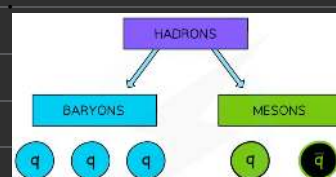
- **electrons and neutrinos are leptons**
- **strong nuclear force does not act on these particles, whilst quarks do**

=> Protons and neutrons are not fundamental particles. They are each made up of three quarks.



=> **Hadrons** are the group of subatomic particles that are made up of quarks. These may be either a:

- **Baryon (3 quarks)**
- **Meson (quark and anti-quark pair)**

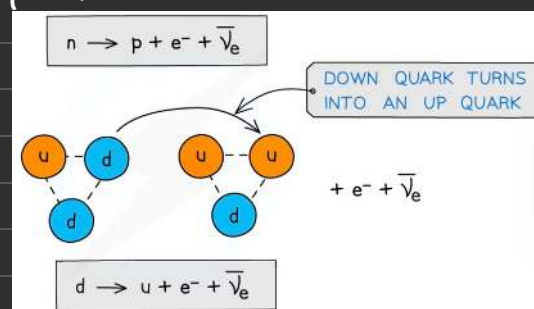


=> All **baryons or meson have integer** (whole number) charges)

=> Combination of quarks and anti-quarks doesn't exist.

=> **Beta decay happens through weak interactions.**

=> **Beta minus decay is when a down quark turns into an up quark**



=> **Beta positive decay is when an up quark turns into a down quark.**

